Riding the Data - Bikeshare Performance Dashboard

Internal report handoff

A Power BI Analysis of Bikeshare Trips, Stations capacity & Weather condition

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What is Capital Bikeshare and what does this data cover

8,000+	800+	4
bikes and e-bikes	docking stations	jurisdictions served

DC · Arlington · Alexandria · Nearby MD and VA jurisdictions

Two sources of data for April-May 2026 analyze:

- Ride data : trip start/end times, stations, and rider types (member/casual).
- Station capacity: dock counts, location and status per station.

It reveals usage patterns like when, where and who rides.

Helping plan growth, correcting pricing and understand weather's effect on demand.

Source: Capital Bikeshare, April-May 2026 · Imported at 22.06.2026 13:58:00

What was built during report

Over the project, in a Power BI pipeline was built queries that joins three sources:

- Ride data (CSV's)
- Station capacity (JSON)
- Daily weather history (CSV)

Unlike the raw files, the pipeline resolves station to trip and weather to trip relationships.

Create calculated fields (trip duration, day of week, duration and duration bins).

And include cross-source comparisons (trips vs. capacity) that aren't possible from any file separatley.

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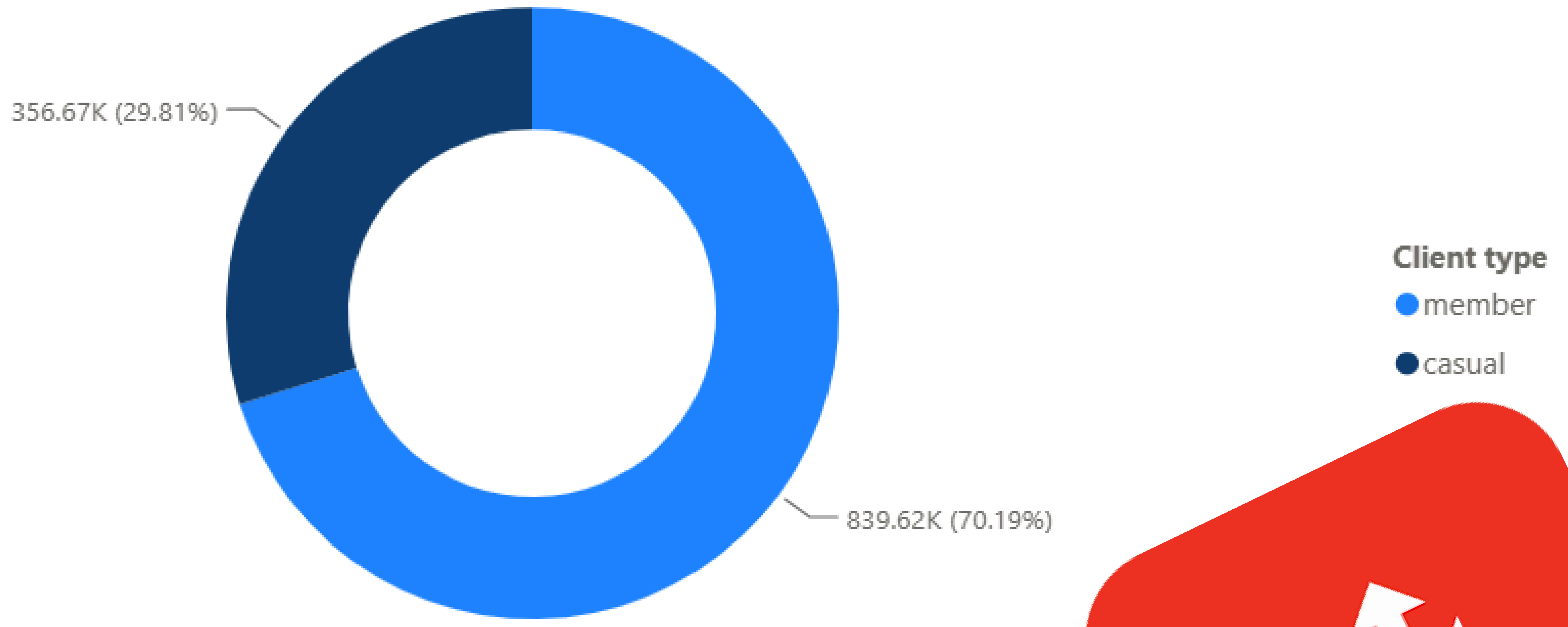
Member Loyalty Drives Volume

April 2026 - May 2026 Date Range	1.2M Quantity of rides	842 Total unique stations	13.24 Average trip duration (min)

Quantity of rides by Client type

Are 70.19% of rides came from members that is nearly 840K trips in April and May 2026. But members ride 3 min shorter average duration compare to casual riders

Members are the backbone of ridership volume but casual riders are the ones taking more time.



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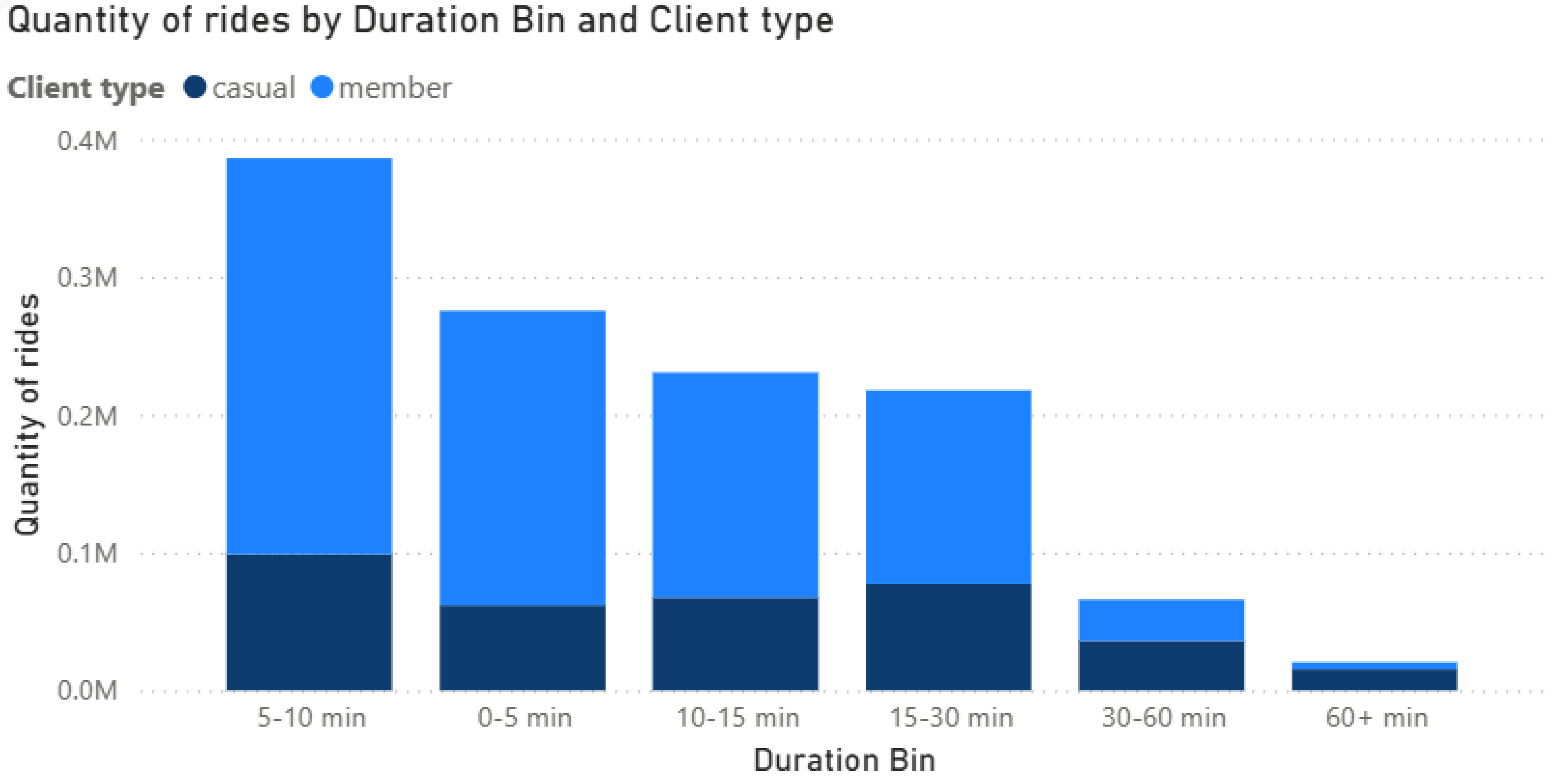
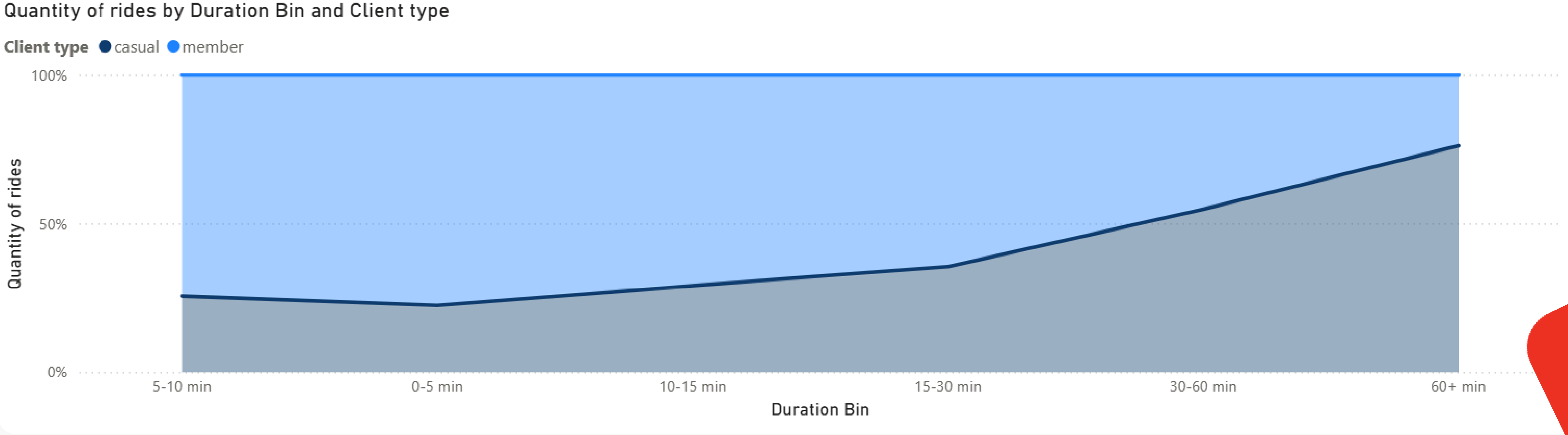
Trip Duration - Members Ride Short, Casual Riders Skew Long



casual	casual
11.09 Median trip duration	18.35 Average trip duration
member	**member**
8.33 Median trip duration	11.08 Average trip duration



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Trip Duration - Members Ride Short, Casual Riders Skew Long

Members show shorter, more consistent trip times.
Median 8.33 min, average 11.08 min.

Casual riders trend longer and more variable.
Median 11.09 min, average 18.35 min, nearly 7 minutes higher on average.

The gap between casual riders' average and median points to a long tail of extended trips pulling the average up.

Casual riders share of rides climbs steadily by duration. That mean long rides are overwhelmingly a casual rider behavior.

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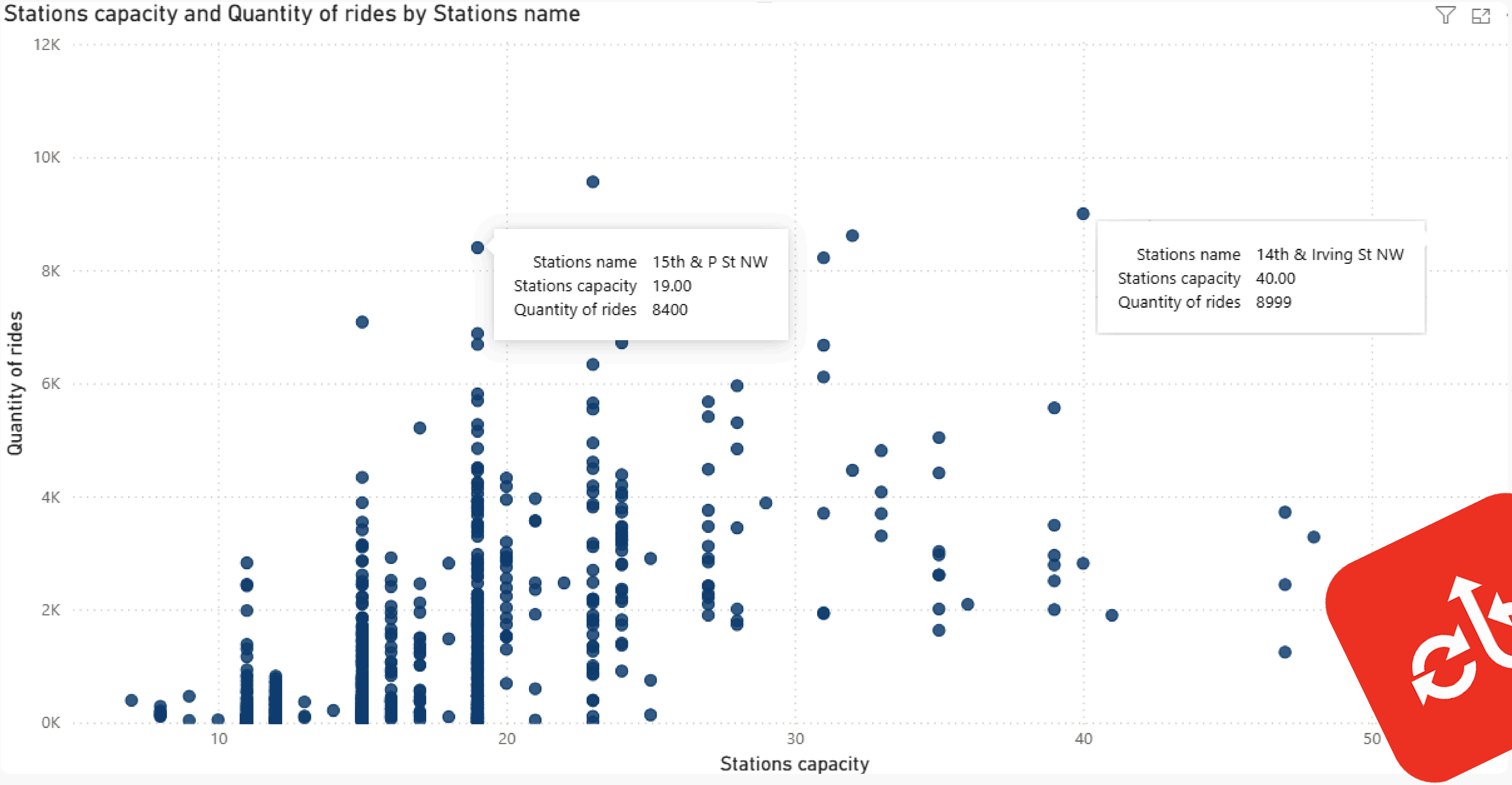
Station Efficiency - Capacity vs. Demand



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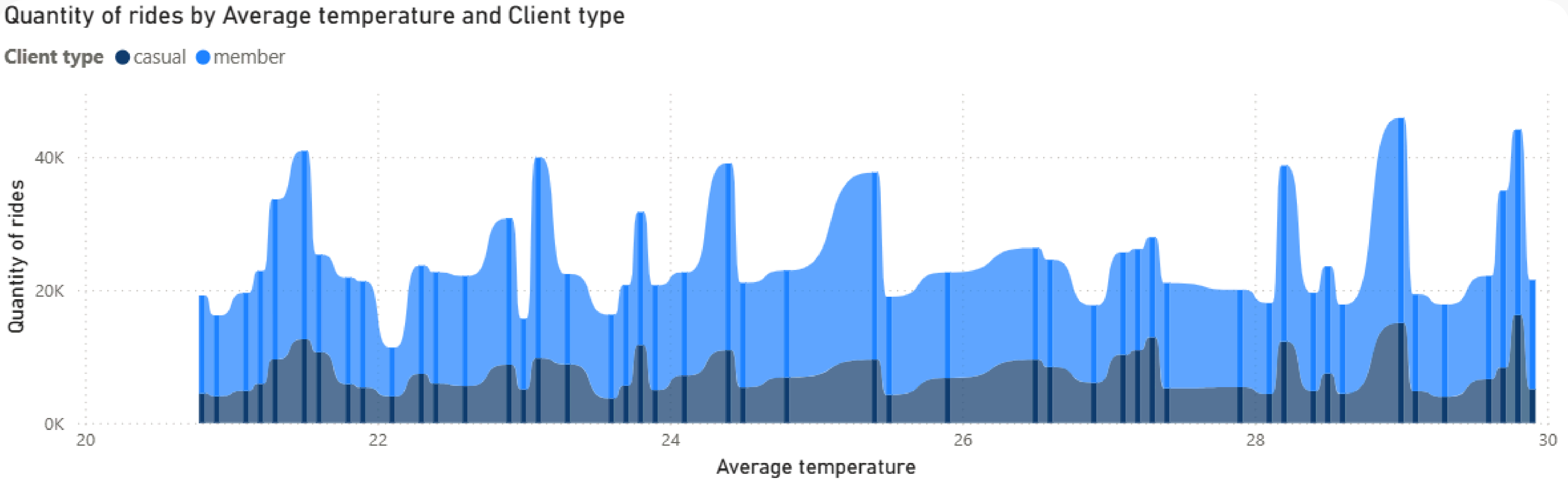
Station Efficiency - Capacity vs. Demand

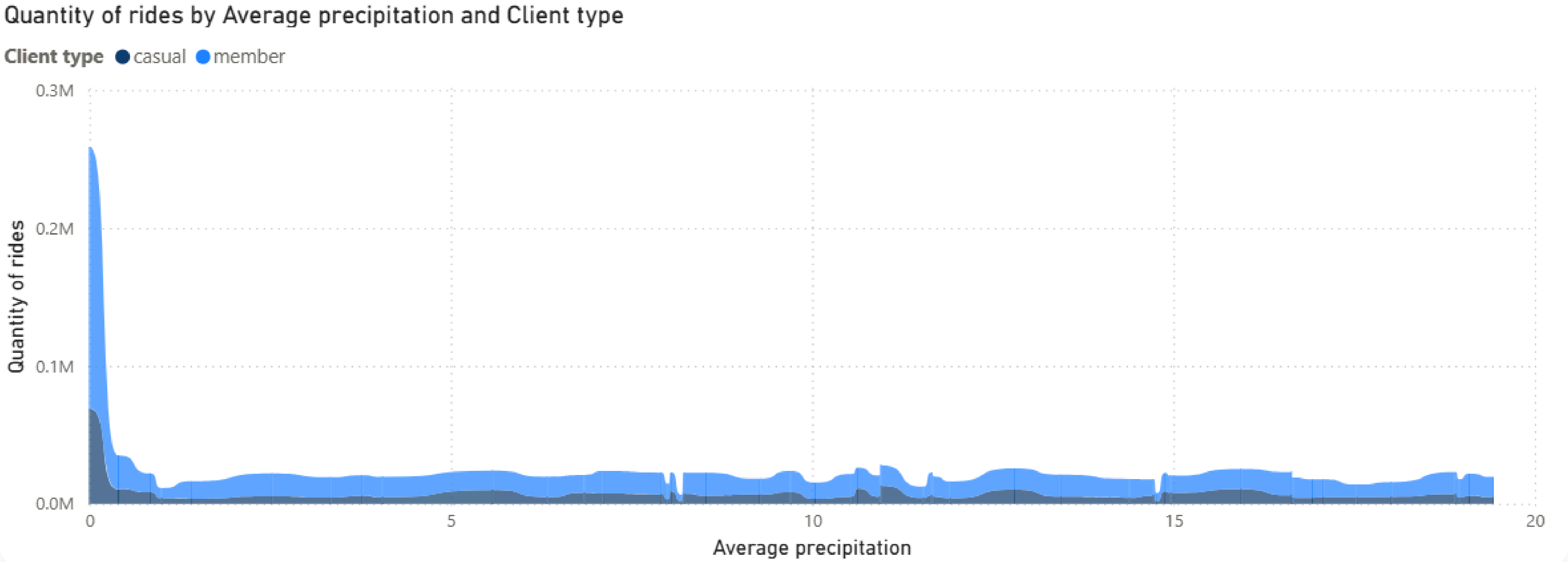
The 15th & P St NW generates nearly twice more rides per dock than 14th & Irving St NW, despite having less than half the capacity.

This points to a potential under-provisioned, high-demand station.
Its may be a strong candidate for capacity expansion, while 14th & Irving may be relatively over-provisioned for its actual usage.

Before making any capacity changes, this needs deeper investigation which including popular routes to confirm the pattern holds.

Ride Volume - Precipitation Matters More Than Temperature





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Ride Volume - Precipitation Matters More Than Temperature

Rides show a strong negative relationship with precipitation.

Rides volume spikes to nearly 0.3 mm rides at near-zero precipitation, then drops sharply and stays flat once precipitation exceeds near by 1mm.

Temperature, by contrast, shows no clear trend.

Rides volume fluctuates between 20K-45K across the entire 20-30° range without a consistent upward or downward pattern.

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Limitations - What the Data Does Not Show

- Two month window only. April-May 2026 captures spring patterns only. Without visibility into summer peak demand, winter lows or full seasonal cycles.

- Station capacity is a snapshot. The dock counts reflect current capacity, not historical changes (rebalancing, temporary closures or expansions during the period aren't tracked).

- Outliers not fully vetted. Extreme trip durations (very short) may include data errors (e.g., undocked bikes) rather than real rides.

- Weather is daily average, not hourly. A rainy morning with a dry afternoon looks identical to a day with light rain all day, which may blur short term ride decision patterns.

- No individual rider data. A trip records show what happened (start/end, duration, member type) but not why. No survey data, trip purpose, or rider demographics.

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Next Steps

Deeper station analysis. Look further into the JSON station capacity data whether current capacity aligns with actual demand patterns identified this week.

Richer weather granularity. Source hourly (not just daily average) weather data to capture into day patterns. Like does a rainy morning suppress commute-time rides even if the afternoon clears up?

Expand the time window. Extend beyond April-May to cover a full seasonal cycle (summer peak, winter lows) and validate whether current findings hold year.

Clean and validate outliers. Investigate extreme trip durations before drawing final conclusions, to separate real rides from likely data errors.

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Call to action

The data shows strong, weather driven ridership patterns and a loyal member base.

The next step is deeper station level and hourly weather data to turn these correlations into actionable operating decisions.

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Thank you!